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RESEARCH-ARTICLE

New measurement of psychological safety for humanoid

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New Measurement of Psychological Safety for Humanoid

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ABSTRACT

In this article, we aim to discover the important factors for determining the psychological safety of humanoids and to develop a new psychological scale to measure the degree of safety quantitatively. To discover the factors that determine the psychological safety of humanoids from an ordinary person's perspective, we studied 919 Japanese, who observed movies of 11 humanoids and then freely described their impressions about what the safety of each humanoid was for them. Five psychologists categorized all of the obtained descriptions into several categories and then used the categories to compose a new psychological scale. Then, 2,624 different Japanese evaluated the same 11 humanoids using the new scale. Factor analysis on the obtained quantitative data revealed six factors of psychological safety: Performance, Humanness, Acceptance, Harmlessness, Toughness, and Agency. Additional analysis revealed that Performance, Acceptance, Harmlessness, and Toughness were the most important factors for determining the psychological safety of general humanoids. The usability of the new scale is discussed.

Categories and Subject Descriptors

H.1.2 [Models and Principles]: User/Machine System.

General Terms

Measurement, Documentation, Experimentation, Security, Human Factors, Standardization.

Keywords

Psychological safety, Humanoid, Psychological scale

1. INTRODUCTION

Recently, Japan have been confronting new circumstances, such as an increasing number of seniors and a decreasing labor force [1], and our society has started to need the services of more robots. For example, in Japan, according to Ministry of Economy, Trade and Industry [2], there will be 4.8 trillion robots by 2025, and more robots will coexist with people in society.

When considering the coexistence of robots with people, it is necessary to reflect on how to ensure the security and safety of people. As for robots that serve as laborers, like industrial robots, one rule is to have fences around robots and keep people out of the robots' work space [3]. However, for robots that are expected to be partners for people, like humanoids, it is important to keep both physical security and psychological safety in mind because the robots will be with people in daily life.

Regarding good psychological impressions of robots, previous studies have revealed several things. For example, a robot's appearance has an effect on humans' impressions of it [4, 5]. Robots' behaviors [6-8] also relate to humans' psychological evaluations of them. As for interactions between people and robots, the timing of robots' utterances [9] or their verbal invitations to people [10] can affect humans' impressions. Moreover, the way in which people encounter robots, whether in real life or through virtual reality, also impacts their psychological impressions [11]. Although there are other studies that deal with psychological impressions of robots, the focus of almost all of the previous studies has been on the engineering technologies proposed in the study, not the psychological mechanism of humans.

As for psychological safety, some of the studies mention it as a key term, but they suggest focusing on the safety of robots along with physical security [6,7]. Japan Robot Association also include psychological safety as a criterion for service robots [12]. However, these definitions about safety have not been sufficiently validated because the questionnaires used to evaluate safety have not been developed for measuring the safety of robots; instead, they look at the issue from the perspective of human personalities [6, 7]. Moreover, the focus is usually on professionals'

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perspectives, not those of ordinary people, who are the potential users of robots. To create psychologically safe robots, it is necessary to discover the concrete factors that compose the psychological safety of robots for ordinary people and to establish a method for quantifying the degree of safety for individual robots. In this article, we focus on humanoids, which are expected to be partners for people in the near future.

More specifically, [13] developed the scale for evaluation of robots. [13] reviewed the previous studies and defined the useful items to evaluate robots in 5 aspects (Anthropomorphism, Animacy, Likeability, Perceived Intelligence, and Perceived Safety). As a result, they decided the 3 items to evaluate perceived for the Perceived Safety. But they translate the items that originally formed as Likert scale to the items formed as SD (semantic differential) method and they don't show the data to investigate the effect of this transformation statistically. Moreover, the number of subjects seems to be relative few to validate the scale satisfactory (36 or fewer participants).

Thus, in this study, we focused on the psychological safety of humanoid robots from ordinary people's general perspectives. That is, we deal with the perceived safety for humanoids from ordinary people's view. We aim to clarify the general factors for evaluating the psychological safety of humanoids, and then develop a new psychological scale that enables us to quantitatively evaluate the degree of safety of various humanoids using common standards. To realize these objectives, we use a social-psychological methodology. In response, we must consider three technical points about this methodology.

The first point concerns the data sampling of people. To avoid the limitation that the obtained data do not reflect the general opinions of people, we must collect the natural impressions about the safety of humanoids from a wide swath of the population. In order to discover the general factors of psychological safety, we must consider certain attributes of the population we are sampling, such as age, sex, and regional characteristics. For example, it has been suggested that people with computer experience or an interest in robots may have a different impression of robots than those who do not have computer experience or an interest in robots [14]. Therefore, in order to obtain the high representativeness of the data in this study, we focused on the Japanese society. We collected data from males and females with a wide range of ages, from their teens to their 70s, according to the distribution of gender and ages in the Japanese population. The study was conducted in three areas in Japan (Tokyo, Osaka, and Fukuoka) that have relatively higher potentials to be robotized cities in the near future. We asked people about experiences of computers and robots according to [14]. We then considered these attributions — age, gender, experiences with or interests in computers and robots — in our investigations.

The second point concerns humanoids as stimulus materials. Several types of humanoids should be used to discover the general factors beyond the types of humanoid robots. It is difficult to define what conditions classify a robot as a humanoid, even in the engineering fields, as wheeled-walking robots are not always regarded as humanoids. However, to clarify the basic factors of safety that are commonly shared by several humanoids, we used 11 different humanoids, including wheeled-walking robots, biped-walking robots, and androids. From an engineering point of view, these types are clearly different in appearance and mechanics; however, this study aims to discover what factors compose the psychological safety of humanoids generally for ordinary people. Therefore, we

included these types of robots and used as many different robots as possible to investigate the differences or similarities in robots with regard to the psychological perspective.

The third point concerns the methodology of analysis. As a first step, we wanted to identify the concrete factors for determining psychological safety, which should be based on the subjective opinions of ordinary people. This means that we needed to ask them about their natural impressions of the safety of humanoids without making any assumptions. Therefore, we only asked them about what they think or feel about the safety of the robots and to describe their impressions in the form of free descriptions in open-ended questionnaires. Although the analysis of qualitative data is technically difficult and requires cumbersome procedures, it is necessary to understand the general factors for ordinary people. As a second step, we wanted to develop a new psychological scale. In this step, it was necessary to use statistical procedures as an objective way of quantitatively discovering the general factors. To conduct factor analysis in the social-psychological field, one usually needs a data sample that is about 10 times larger than the number of items on the scale in order to obtain valid results, although a larger number is better statistically. Based on our previous study, we assumed the scale would be around 40 items [15]. Thus, we collected more than 400 people to develop the new scale to make it a general scale.

Therefore, in this study, we aimed to clarify the concrete factors that affect ordinary people's ideas regarding the psychological safety of humanoids by collecting both qualitative and quantitative data from a wide swath of the population and using 11 different humanoid robots. This social-psychological approach can help develop a new scale for evaluating the psychological safety of humanoids generally.

2. Study 1

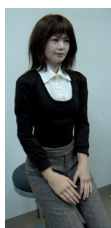
We used the same movies of humanoids, way of collecting data, and procedures for conducting the presentation experiments in both studies 1 and 2. Therefore, we describe the method for both studies here. In Study 1, we aimed to collect the natural thoughts or feelings about the safety of humanoids for ordinary people in order to discover the concrete factors of psychological safety from qualitative data.

2.1 METHOD

2.1.1 Humanoid Robots and Movies

Figure 1 shows the 11 robots that were presented as stimuli in this study. To obtain general opinions, it is necessary to place the humanoids in as many contexts as possible. Using real robots limits the context to just the time and place that people experience the robot, and it is very difficult to activate all 11 robots to show them to many people for a sufficient amount of time and in sufficient contexts. Therefore, we showed movies of the robots. Movies can show a variety of contexts at once, which is necessary to obtain various opinions, as long as the movies provide a variety of scenes or natural sounds.

Thus, the movies were made to show a wide range of possible situations, including robots alone and interacting with people, that cover the main abilities of each robot as much as possible. We focused on situations people face in their daily lives; therefore, we did not include the background of the operations but mainly focused on the actions of robots interacting with users, not being operated by people.

Wheeled-walking	 Robovie[16]	 wakamaru[17]	 enon[18]
	 ASIMO[19]	 HRP-2 [20]	 HRP-4C [21][22]
	 CB2[249]	 Repliee Q2[25]	 Geminoid F[27]

We explained to the developers of each robot the aim of the movies and asked them to provide movies that showcased the main abilities of their robots. We then edited the scenes to show all of the robots' abilities or activities. After editing, we asked the developers to again check the movies from a professional point of view to confirm that there were no missing scenes and no bias had been inserted. After several adjustments, we created final versions of all of the movies, except for one for enon, because we could not contact the developer. Every movie came with sound and was about four minutes long.

2.1.2 Participants

The population provided by the Ministry of Internal Affairs and Communications was used to calculate the number of data samples. Empirical social-psychological studies use 15-35 data samples for one experimental condition; therefore, we used 25 participants for the smallest sample generation and calculated the numbers for other generations according to the Japanese population. The total number of participants for each group in Study 1 was 225.

Showing all 11 robots to participants could induce a psychological overload and prevent us from obtaining data about their natural impressions. Therefore, we divided the 11 robots into four groups containing even numbers of each type of robots (Table 1). Each group had 225 participants, so the number of total participants for Study 1 was 900. (Participants in Study 2 are described in the appropriate section below.)

Table 1. Number of participants in each experiment

	Robots	Tokyo	Osaka	Fukuoka	Total
1	Robovie, ASIMO, Repliee Q2	75	75	75	225
2	HRP-2, Geminoid F, Neony	75	75	75	225
3	HRP-4C, wakamaru, CB2	75	75	75	225
4	enon, Geminoid	75	75	75	225
	total	300	300	300	900

A marketing office that has dealt with various surveys for public offices and universities conducted the sampling of participants and surveys. In order to produce a fair collection of participants, they did not use limited sampling methods such as online surveys. It is much easier and less expensive to use online surveys or people who are already listed as participants for the marketing office. However, only using such sampling methods could produce data that reflect only the opinions of the limited number of people who use the Internet or belong to one office. Therefore, for gathering participants, the marketing office used both online monitors (about 20%) and newspaper advertising (about 80%) to produce a sample population that was representative of Japan. Marketing offices in Japan are not allowed to let people under age 14 age participate in studies, so the group of teenagers used in the study ranged in age from 15 to 19.

2.1.3 Procedure

As Table 1 shows, 75 people were selected for each experiment and location. The 75 people were divided again into three groups of 25. Then, each group of 25 participated in the experiment at least once, and the experiments were repeated.

The facilitator distributed the open-ended questionnaires and instructed the participants to watch a movie and then describe their impressions. As for the impressions, we focused on the safety of the robots, but security is a similar concept, so we asked participants about both safety and security. If we found different results for security and safety, it would be necessary to develop identical scales for both security and safety as psychological impressions, although we assumed that security was physical and safety was psychological. If we found no difference, we could use one scale. The order of the questions about security and safety were randomized for balance.

After the participants watched the movie, they described their impressions in the questionnaire, which asked them to describe what they thought or felt about the safety and security of the humanoid, being as detailed as possible. After one movie, they watched the movie about the second robot and repeated this sequence. Three surveys were repeated three times for three robots, but only one survey was repeated twice for the two robots enon and Geminoid. The order in which the robot movies were shown was randomized for balance.

We used one DVD player (Pioneer DV220V), one projector (Sanyo LP-XU75), two speakers (Bose 101MM), one amplifier (Bose 1705II), and one screen (100 inches, or 2 m × 1.5 m) to display the movies.

After describing all of the assigned robots, the participants completed three questionnaires about their interactions with computers: frequency of computer use = (1) not at all, (2) a few times a year, (3) a few times a month, (4) a few times a week, or (5) almost every day; their interest in robots (how many times

they watch science-fiction movies in a year); and how many times they have seen real humanoids. We used these questions [14] to show how these factors affect people's impressions of robots.

2.2 RESULTS

2.2.1 Obtained samples and analysis of descriptions

As Table 2 shows, we eventually obtained more data samples than first expected. The number of participants in their 70s was slightly less than their actual representation in the Japanese population. In fact, we were able to secure the expected number of participants in their 70s, but it was sometimes difficult for them to come to the conference room. However, the smaller number of samples was not significant enough to greatly affect the representativeness of the population.

Table 2. Obtained data and excess and deficiency.

Age	Obtained data		Excess and deficiency	
	Male	Female	Male	Female
15-19	53	49	1	1
20-29	61	64	1	4
30-39	77	82	1	6
40-49	70	74	2	6
50-59	72	73	4	5
60-69	74	77	2	1
70-79	42	51	-6	-9
Total	449	470	5	14

Table 3. The number of descriptions about both safety or security in each robot

	Robot	Safety	Security	Total
1	Robovie	861	791	1,652
	ASIMO	928	880	1,808
	Repliee Q2	900	753	1,653
2	HRP-2	989	972	1,961
	Geminoid F	1,027	867	1,894
	Neony	966	924	1,890
3	HRP-4C	802	714	1,516
	Wakamaru	834	776	1,610
	CB2	787	692	1,479
4	enon	1,181	1,134	2,315
	Geminoid	1,065	1,016	2,081
Total		10,340	9,519	19,859

The study was divided into four surveys; therefore, it was necessary to confirm that the surveys did not differ in terms of sample quality. In particular, the study focused on ordinary people's impressions of humanoid robots. We conducted one-way ANOVA for the questionnaire of [14] about computer use and interest in robots. We did not find a significant difference in the surveys for all three questions. This means that the four sample groups did not greatly differ in terms of their interactions with computers and robots, which could affect their impressions of humanoid robots. Therefore, the sample groups could be considered the same in the following analyses.

As Table 3 shows, we obtained a total of 19,859 descriptions (safety: 10,340; security: 9,519). All of the descriptions needed to be categorized into groups reflecting distinct psychological meanings for both safety and security. To analyze the qualitative data, it was important for the researchers to remain objective during the categorization process. Therefore, this study engaged

five psychologists, who agreed to remain objective, for the analysis.

First, each description was written on a small card that differed in color for each robot. The five psychologists picked the cards randomly and checked the contents of the descriptions. After checking a few hundred cards, they discussed which psychological meanings appeared; then, they formed a few trial categories and named them. They then assigned the cards to the trial categories. As the card checking continued, the categories were modified. Throughout the analysis, the psychologists repeatedly checked the descriptions, discussed the categories, and assigned the cards to them until all of the cards had been assigned. They also conferred about their choices in order to maintain mutual agreement with one another. The analysis was done separately for both safety and security to compare the results of the obtained categories

2.2.2 Obtained categories

After categorization, the following 10 groups were obtained: Performance, Utility, Acceptance, Harmlessness, Motion, Design, Humanness, Robotness, Synonymity, and Other (Tables 4 and 5).

Table 4. Revealed categories and contents.

Category	Contents
Performance	high performance like accurately performing a task as directed or having an error-free system so it could not go out of control; also limited performance so it could not exceed human abilities
Utility	high usefulness as an instrument to support people, high cognitive abilities, high intelligence
Acceptance	high familiarity, comfort, mental healing
Harmlessness	ability to not fall down, keep its balance, not clash with people or its surroundings, and adjust its physical power
Motion	slow and smooth motions, ability to not act surprisingly actively
Design	round shape, not too heavy, soft quality of materials, soft color
Humanness	communicative; human likeness; human-like voice, motion, and appearance
Robotness	not human-like; robot-like voice, motion, and appearance
Synonymity	descriptions that mention "impressions are equal for both safety and security"
Other	has a social security system to keep information private and not allow it to be misapplied, etc.

Table 5. Number of descriptions in each category.

Category	Safety	Security	Total
Performance	1,049	1,483	2,532
Utility	1,589	1,742	3,331
Acceptance	1,082	493	1,575
Harmlessness	611	1,645	2,256
Motion	886	674	1,560
Design	1,319	871	2,190
Humanness	1,940	622	2,562
Robotness	484	224	708
Synonymity	856	823	1,679
Other	524	942	1,466
Total	10,340	9,519	19,859

As a result, we found similar categories for both safety and security. The descriptions were also sufficiently similar that we could compose just one category and conclude that safety and

security have the same meaning in regard to humanoids. Therefore, we deal with these descriptions in terms of overall psychological safety. “Other” descriptions were gathered into small categories, but they did not fit into other categories very well, and there were not enough descriptions to make a single category. Therefore, the following analyses focused on the nine main categories.

2.3 DISCUSSION

In Study 1, we aimed to identify the concrete factors of psychological safety. Over 900 Japanese, from teenagers to people in their 70s, participated in the study. We identified eight factors of safety, as well as two additional categories that were not core factors, Synonymity and Other.

Relative to the other categories, Performance and Utility have the most descriptions. It seems people tend to think a high level of technology is related to safety or security. Humanness is also a major category, whereas Robotness has few descriptions. However, Robotness has almost the opposite meaning of Humanness, and totally human-like qualities seem to be an important factor in how humans regard the safety of humanoids. Relatively few descriptions are categorized in Acceptance, but this category is clearly independent from the other categories in terms of its contents, such as mental comfort. Harmlessness, Motion, and Design focus on the idea that humanoids do no harm to people.

This study used movies to present robots and there is room to investigate whether the obtained perspectives about safety or security are different from them in situations to see real robots or not. In another study, we investigated basic perspectives to evaluate general impressions of humanoids using movies as same as this study. And we conducted an experiment to see the differences by using real ASIMO in the situations that ASIMO introduced itself to each 30 ordinal young adults who didn’t have special knowledge about the robot like participants in this study. As a result, they tended to use fewer and limited perspectives especially related to the interactions than the revealed categories through observing movies. It seems that the situation with a real robot could identify the context, therefore, it would be easy to focus on more specific perspectives rather than broader and many perspectives. That is, the perspectives revealed in this study that showed movies of broad contexts as much as possible could obtain general perspectives about safety or security than the limited real contexts, whereas the future research is necessary to clarify more details about this point.

Next, to develop the psychological scale, we considered the meanings of each category and reworded the descriptions into items for a close-ended questionnaire. Based on the descriptions, we created 48 items (each category had five to 10 items). We developed the items according to the contents of descriptions that are dominant in the all of obtained data. Mainly, the major descriptions in each category were used and reworded a little to make them appropriate form as psychological scale. In Study 2, we refined this scale using statistical analysis.

3. Study 2

In this study, we aimed to analyze the new scale using an objective statistical method. We asked different participants from Study 1 to evaluate a humanoid using the new scale to obtain quantitative data that would reveal each robot’s degree of safety. By means of factor analysis, we clarified the factor structure of the psychological safety of humanoid. It also revealed the reliability (Cronbach’s alpha) of the psychological scale.

3.1 METHOD

We used the same movies of the 11 humanoids in Study 2 as in Study 1. However, we collected different participants for Study 2. As for the number of participants, to conduct empirical factor analysis in the social-psychological field, one needs a data sample that is about 10 times larger than the number of items on the scale. This required a data sample of about 480 for this study. In this article, we focused on the basic factor structure across all of the humanoids, and we processed the data for all 11 humanoids using factor analysis. Therefore, we eventually obtained enough data to meet this minimum number, but analysis of each robot’s factor structure (which was not a main issue in this article) requires about 10 times as much data. In addition, it was necessary to maintain the same population distribution as we used in Study 1 (Figure 2). Therefore, we doubled the number of participants we used in Study 1, which resulted in 450 participants for one presentation experiment. In this study, we divided the 11 humanoids into six groups, so as not to overload the participants psychologically. The total number of participants was 2,700. We presented the robots in a random order for balance.

We used the same procedure in Study 2 as in Study 1, including the location of the experiments and the instruments used for showing the movies. We also used the questionnaire of close-ended items that was developed in Study 1. The questionnaire contained 48 items, to which participants responded on a seven-point Likert scale, which ranged from 1 (strongly disagree) to 7 (strongly agree). To investigate whether the developed items could fully measure the robots’ psychological safety, we added an additional level of analysis by adding six items that were made to measure safety directly: “I feel safe around this robot,” “I feel relieved with this robot,” “I feel at ease with this robot,” “I become anxious with this robot,” “I feel messed up with this robot,” and “I get stressed with this robot.” The last three items were reversed, and all six items were combined to create a score for “tentative safety,” which was based only on words directly related to safety. After factor analysis, we investigated the correlation between scores based on the obtained factor structure and used the tentative safety score as an easy check that the scores based on the other factors could fully measure the robots’ psychological safety.

3.2 RESULTS

3.2.1 Obtained samples

As Table 6 shows, we eventually obtained fewer data samples than we first expected. However, for this article, we processed all of the data obtained from the six experiments, which totaled 2,624, a satisfactory number.

Table 6. Obtained data and excess and deficiency.

Age	Obtained data		Excess and deficiency	
	Male	Female	Male	Female
15-19	131	149	-25	5
20-29	196	183	16	3
30-39	230	233	2	5
40-49	206	216	2	12
50-59	189	195	-15	-9
60-69	209	209	-7	-19
70-79	136	142	-8	-38
Total	1,297	1,327	-35	-41

Table 7. Example of items and Factor structure and reliability of the new psychological scale for safety of humanoid

	1	2	3	4	5	6
This robot seems to respond appropriately to my question.	0.76	0.15	-0.02	0.05	0.02	-0.11
This robot seems to be able to understand what humans say.	0.75	0.14	-0.07	0.06	0.00	-0.03
I think this robot can recognize its surroundings.	0.70	-0.08	-0.01	-0.06	-0.07	0.03
This robot seems to work according to my instructions.	0.66	-0.05	-0.04	0.08	0.02	-0.24
Its figure is robot-like.	0.14	-0.90	0.08	0.05	0.13	0.04
Its appearance is similar to that of humans.	0.10	0.83	0.01	0.04	0.14	-0.13
This robot does robot-like motions.	0.12	-0.76	0.01	0.11	0.21	-0.09
It makes me self-conscious that this is a robot.	0.16	-0.74	0.07	0.08	0.13	-0.17
I am familiar with this appearance.	-0.03	0.01	0.80	-0.09	-0.03	-0.01
This robot moves in a friendly way.	0.10	-0.01	0.70	-0.01	0.00	0.07
This robot seems to give me mental healing.	0.00	-0.09	0.67	0.07	0.04	0.29
I think I can have a cozy atmosphere with this robot.	0.12	-0.17	0.63	0.08	-0.07	0.16
This robot doesn't seem to do injury to a human's body.	-0.04	-0.01	0.01	0.68	-0.09	-0.03
This robot doesn't seem to go out of control.	0.10	0.02	-0.11	0.65	-0.22	0.08
I do not think this robot can exceed a human's ability.	-0.24	-0.01	-0.02	0.46	0.13	-0.13
This robot seems to produce improper operating signals.	-0.24	-0.09	0.07	-0.14	0.49	-0.17
This robot seems to break down easily.	-0.30	0.00	0.06	-0.01	0.45	-0.20
This robot looks like it falls down on the ground easily.	-0.26	0.14	-0.02	0.08	0.41	-0.10
This robot seems to have an independent will like humans.	0.25	0.13	-0.04	-0.16	0.09	0.55
This robot doesn't seem to have emotions like humans.	-0.11	-0.17	-0.10	0.11	0.17	-0.46
Interfactor correlation matrix						
Performance	1.00	0.29	0.65	0.31	-0.06	0.34
Humanness	0.29	1.00	0.22	0.00	0.18	0.37
Acceptance	0.65	0.22	1.00	0.41	-0.11	0.21
Harmlessness	0.31	0.00	0.41	1.00	-0.28	-0.10
Toughness	-0.06	0.18	-0.11	-0.28	1.00	0.30
Agency	0.34	0.37	0.21	-0.10	0.30	1.00
α	.88	.89	.84	.60	.60	.65

3.2.2 Factor analysis

We conducted an exploratory factor analysis with promax rotation on scales. Analysis on the item level according to Kaiser's criterion (eigenvalues > 1.0) yielded nine factors. Inspection of the scree plot revealed fewer factors. However, overall interpretive significance revealed that a model containing six factors was more appropriate. This result accounted for 48% of the variance in the data. In order to create the final form of the scale, we selected 33 items that loaded above .40 on each factor (and did not load above .30 on any other factor).

Table 7 lists examples of items, factor loadings, interfactor correlations, and the reliability of each subscale. The first factor, Performance, included a high level of performance, such as appropriately working as instructed or having useful actions when people needed them. The second factor, Humanness, captures the human-like (as opposed to robot-like) qualities of the robot in terms of appearance and motions. The third factor, Acceptance, clearly reflects a feeling of comfort or familiarity with the robot. The fourth factor, Harmlessness, captures the sense of the robot's obedience, such as not harming people or not going out of control. The fifth factor, Toughness, concerns robots not malfunctioning or breaking down easily. The last factor, Agency, mentions the robot's human-like will or emotions.

In Study 1, we identified eight categories: Performance, Utility, Acceptance, Harmlessness, Motion, Design, Humanness, and Robotness. In contrast, this Study 2 revealed six factors. The data suggest that Performance and Utility can be combined into a single factor: Performance. Further, Design (like an item about a round-shaped design) and Motion (like an item about amiable motions) can also be combined into a single factor: Acceptance. Part of the Performance category (like an item about limited ability) was combined to become a factor of Harmlessness. Then, a part of Harmlessness (like not falling down) and Performance

(like not going out of control) were combined to create Toughness. Humanness remained the same, and Robotness became Agency.

In Study 1, we categorized the cards based on meanings and wordings with as much detail as possible. The results of factor analysis showed a response pattern to evaluate. Therefore, the categories in Study 1 and the factors in Study 2 are slightly different. However, these factors each have a clear meaning and seem to reflect the structure of humans' psychological reactions toward the safety of humanoids. As for reliability (alpha score), Performance, Humanness, and Acceptance show a high reliability as a psychological scale, while in comparison, Harmlessness, Toughness, and Agency show a relatively low reliability.

3.2.3 Correlation check

Factor analysis revealed that there are six factors of humans' perceptions about the psychological safety of humanoids. In this section, we investigate whether these factors can account for the concept of safety. Because we collected unstructured, free descriptions in Study 1, it is possible there are more or fewer important factors than these six. We calculated the average score of each factor and the score of tentative safety (i.e., the six items that asked about safety directly), and the correlations are shown in Table 8.

As Table 8 shows, Acceptance, Performance, and Toughness are strongly correlated to the tentative safety score. That is, these three factors are important factors for explaining the psychological safety of humanoids. Harmlessness also shows a positive correlation. However, Agency shows a low correlation, and Humanness is not significantly correlated. These two factors do not seem to relate to psychological safety very strongly. To investigate further details, we conducted a correlation analysis of the 11 humanoids.

Table 8. Table captions should be placed above the table

		1	2	3	4	5	6
1	Performance	1	.28**	.62**	.15**	.44**	.416**
2	Humanness	.28**	1	.28**	-.08**	.03*	.48**
3	Acceptance	.62**	.28**	1	.20**	.31**	.30**
4	Harmlessness	.15**	-.08**	.20**	1	.18**	-.18**
5	Toughness	.44**	.03*	.31**	.18**	1	.14**
6	Agency	.42**	.48**	.30**	-.18**	.14**	1
7	Tentative Safety	.58**	0.02	.79**	.28**	.40**	.19**

** $p < .01$, * $p < .05$

Table 9 shows the correlation between the score of tentative safety and the scores of the six factors for each of the 11 humanoids. The results show that almost all of the robots show a significant positive correlation. Acceptance, Performance, and Toughness are especially strongly correlated with tentative safety for every humanoid. However, Humanness and Agency show a low correlation again. As for the Geminoid humanoid, the correlation is not significant between tentative safety and Humanness. This seems to show that being more human-like is not significantly related to psychological safety for some robots.

Table 9. Correlation between the score of tentative safety and the scores of the six factors

	Performance	Humanness	Acceptance
Robovie	.613**	.263**	.787**
Wakamaru	.606**	.194**	.827**
Enon	.621**	.205**	.804**
ASIMO	.643**	.280**	.830**
HRP-2	.594**	.248**	.784**
HRP-4C	.534**	.219**	.793**
Neonv	.520**	.355**	.816**
CB2	.620**	.375**	.842**
Geminoid	.556**	0.051	.759**
Geminoid F	.624**	.306**	.797**
Repliee Q2	.585**	.290**	.818**
	Harmlessness	Toughness	Agency
Robovie	.266**	.411**	.317**
Wakamaru	.217**	.293**	.195**
Enon	.207**	.397**	.282**
ASIMO	.336**	.463**	.245**
HRP-2	.276**	.369**	.206**
HRP-4C	.303**	.348**	.279**
Neonv	.259**	.300**	.224**
CB2	.292**	.469**	.293**
Geminoid	.309**	.390**	.138**
Geminoid F	.167**	.348**	.246**
Repliee Q2	.341**	.343**	.249**

** $p < .01$, * $p < .05$

3.3 DISCUSSION

In Study 2, we aimed to collect quantitative data based on the items developed in Study 1 to investigate the factor structure and reliability of our scale. In addition, correlation analysis was conducted to investigate whether all of the obtained factors are importantly related to psychological safety.

Factor analysis revealed six distinct factors. Some factors of reliability did not show satisfyingly high scores, as one of useful way, it could be helpful to use only high-loaded items. Correlation analysis between the score of tentative safety and the score of each factor revealed that Acceptance, Performance, and Toughness (and relatively Harmlessness) are core factors of psychological safety, whereas Humanness and Agency are not as important.

4. CONCLUSION

In this article, we aimed to identify the concrete factors of psychological safety for humanoids and to develop a new scale that could quantitatively evaluate the degree of safety of specific humanoids. In Study 1, we collected the free descriptions of people's subjective impressions about the safety of humanoids to clarify what people think about the safety of humanoids. Study 2 revealed the factor structure of psychological safety using evaluation data based on items that were developed according to the descriptions and categories obtained and clarified in Study 1. We found six factors of psychological safety: Performance, Humanness, Acceptance, Harmlessness, Toughness, and Agency. Correlation analysis revealed that Acceptance, Performance, and Toughness are core factors of psychological safety for humanoids, and Harmlessness also seems to important relatively. However, Humanness and Agency are not as important.

Previous studies have dealt with the safety of robots, but they have only used psychological scales that were developed to evaluate humans' personality or impressions or from professional point of view. This study identified the structure of psychological safety from potential user's view and developed a new psychological scale oriented to humanoids, especially to quantify their degree of safety.

This research used 11 different humanoids that are currently the most advanced robots available. It also collected data from a wide swath of the population in Japan, ranging in age from teenagers to people in their 70s based on the country's overall population distribution. We used both qualitative and quantitative methods to mold subjective psychology into a more objective structure. Therefore, the scale we developed has a high potential of usefulness for other humanoids and to be used by any person in Japan. In addition, the psychological scale is very easy to use as a questionnaire. It can also be used in its full-length version or shortened to include only high-loaded items. Moreover, according to the aim of this research, it is possible to use only some of the factors. This study widens the possibility of evaluating humanoids psychologically using common standards of safety.

In the future, it is necessary to compare other humanoids with the humanoids used in this study, as well as how people with similar characteristics, such as age and gender, evaluate the safety of humanoids.

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